Optimising for Energy Efficiency and Performance in Machine Learning

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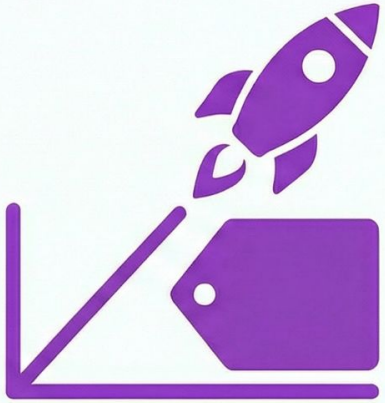
CAIN 2026

The Escalating Cost of AI Scaling

Training GPT-4 cost more than $100 million.

Training BLOOM emits 24.7 tonnes of CO2eq.

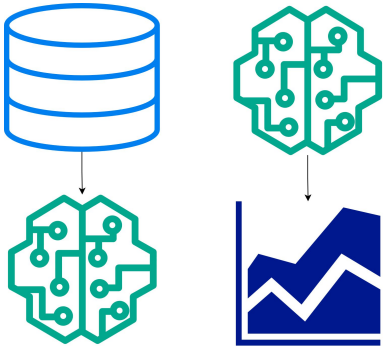


Frontier training costs to exceed $1 billion by 2027.

NotebookLM

Ben Cottier et al. “The rising costs of training frontier AI models”. arXiv: 2405.21015 [cs.CY], 2025
Alexandra Luccioni et al. “Estimating the Carbon Footprint of BLOOM, a 176B Parameter Language Model”. Journal of Machine Learning Research, 2023

Existing gaps



Training → Inference

ML Energy Scaling

Actionable Feedback

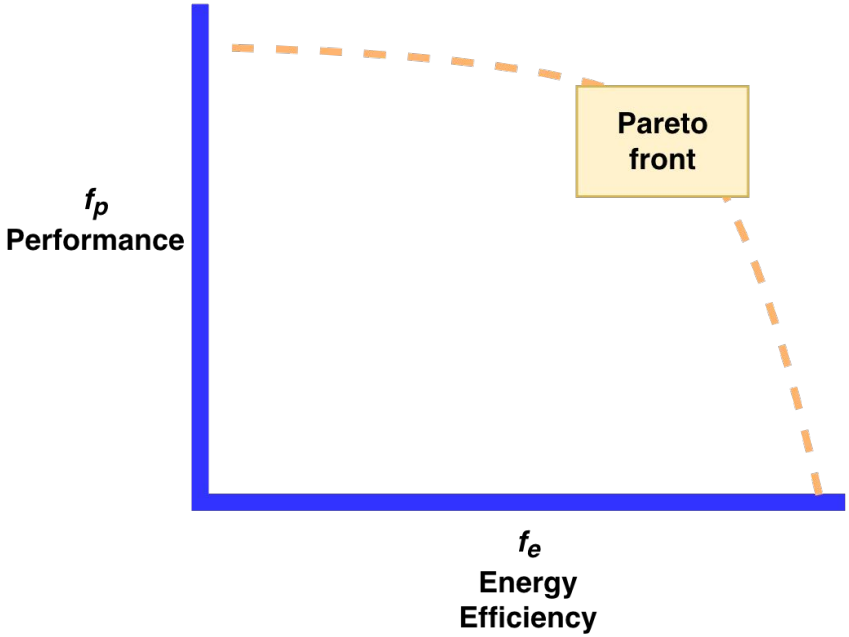
"Our customers are taking to Machine Learning in a big way ... inferencing can account for up to 90% of the cost of their machine learning work."

AWS

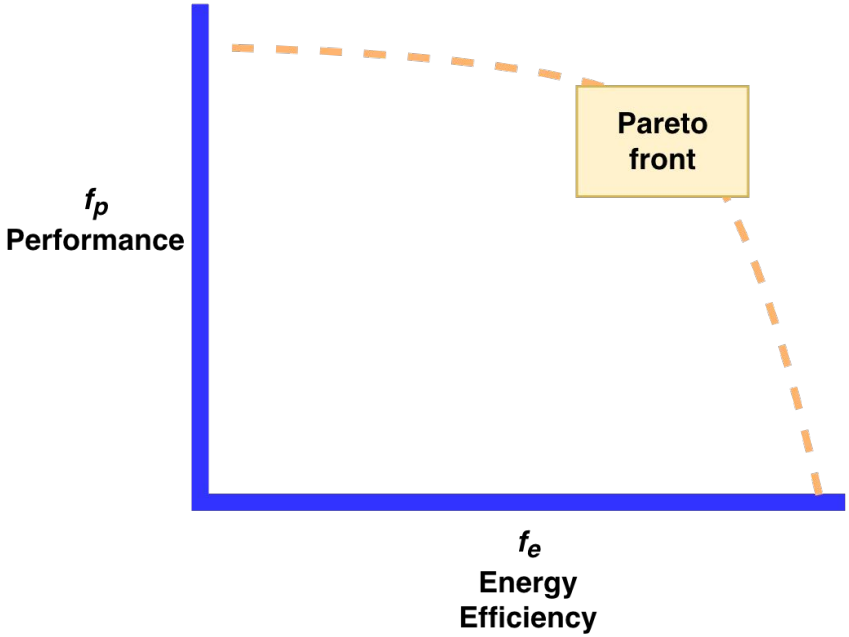
Jeff Barr. 2019. Amazon EC2 Update – Inf1 Instances with AWS Inferentia Chips for High Performance Cost-Effective Inferencing.

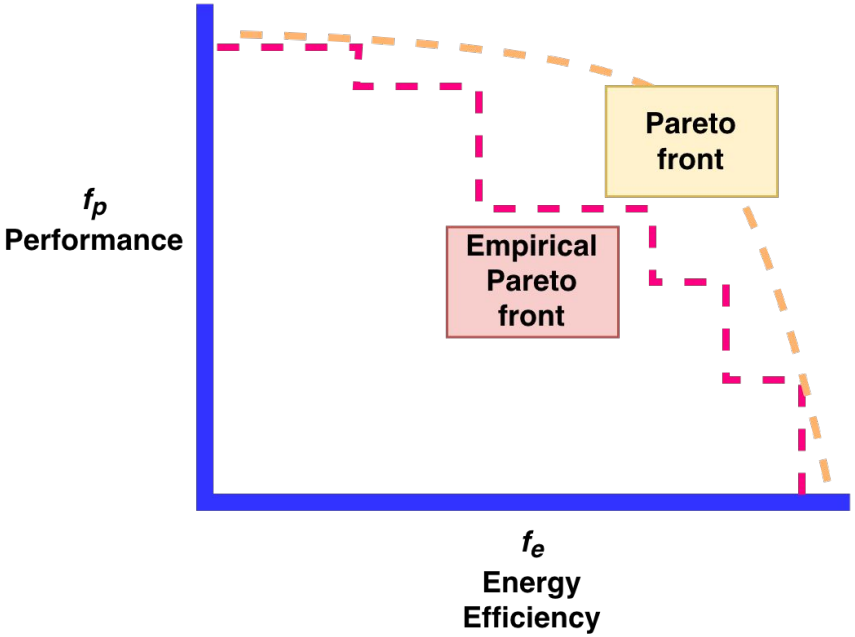
RQ: How to quantify a trade-off between performance and energy efficiency in ML inference?

Multi-objective optimisation

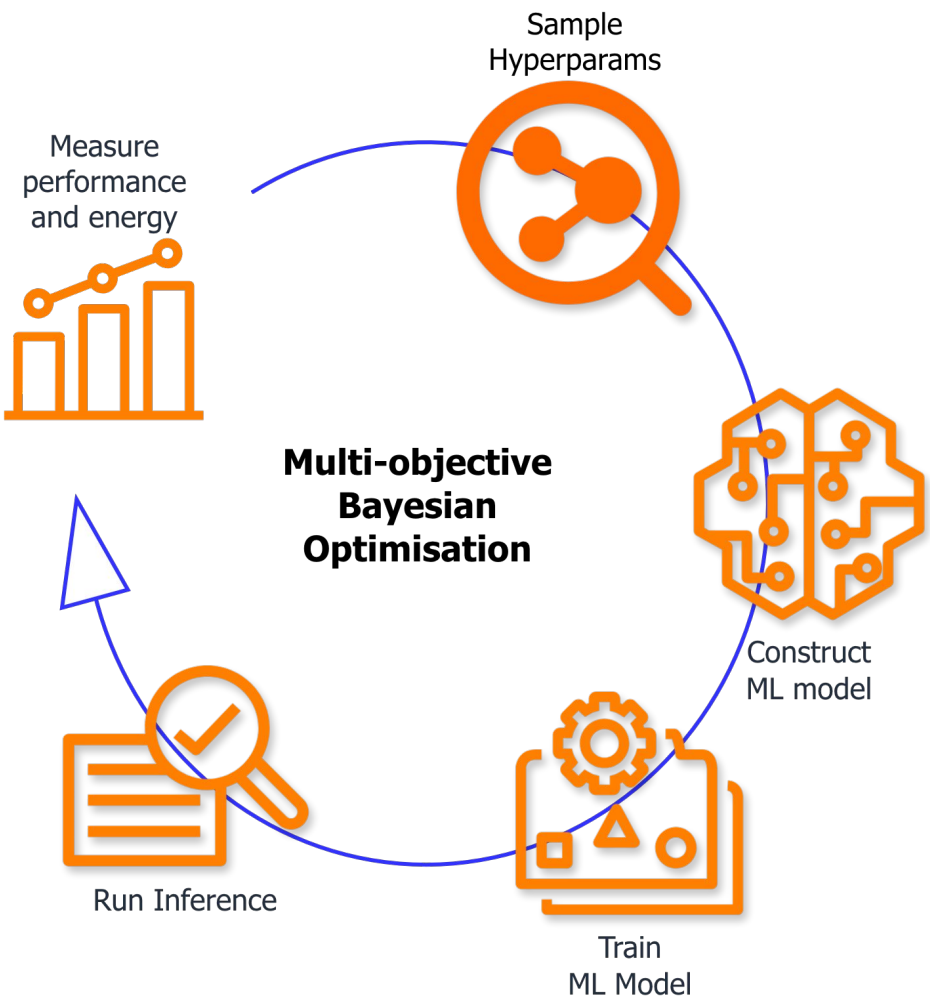


Multi-objective optimisation





Multi-objective optimisation



What is being optimised?

$h_1, h_2, \ldots, h_n$ - hyperparameters

$f(h_1, h_2, \ldots, h_n) =$

[Model → Training → Inference → Measurements] =

(Performance, Energy)

Bayesian optimisation

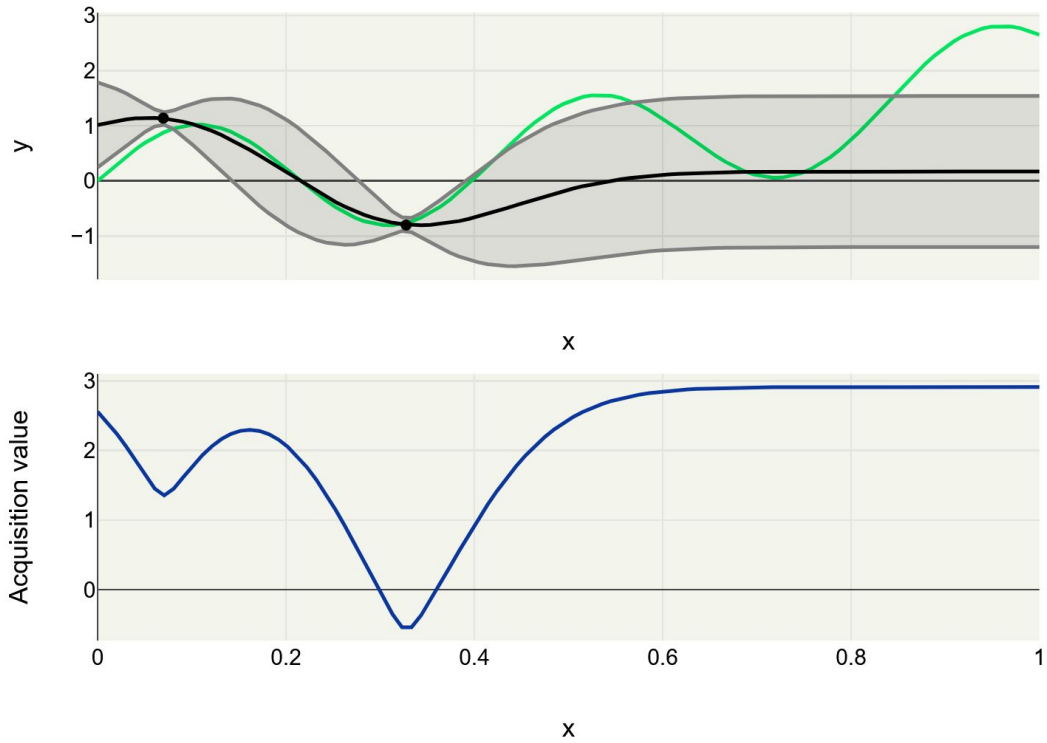
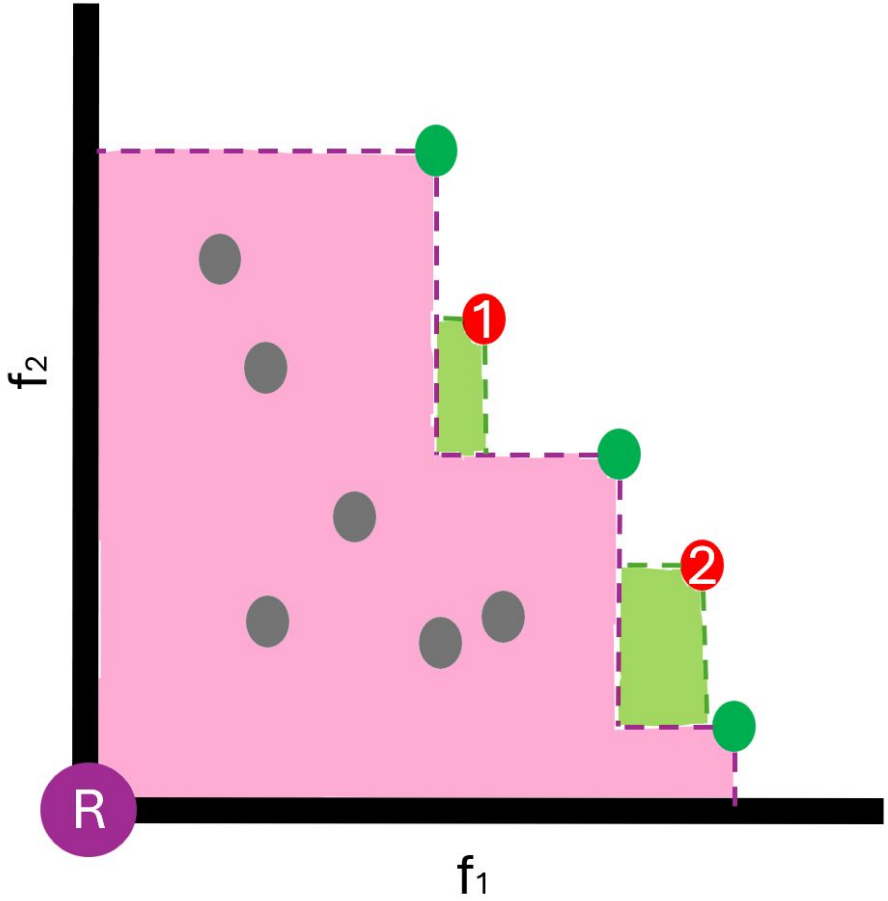


Image source: "Bayesian Optimization and Active Learning Cookbook", Sam Lishak, Physics X, 2025

Multiple objectives: hypervolume



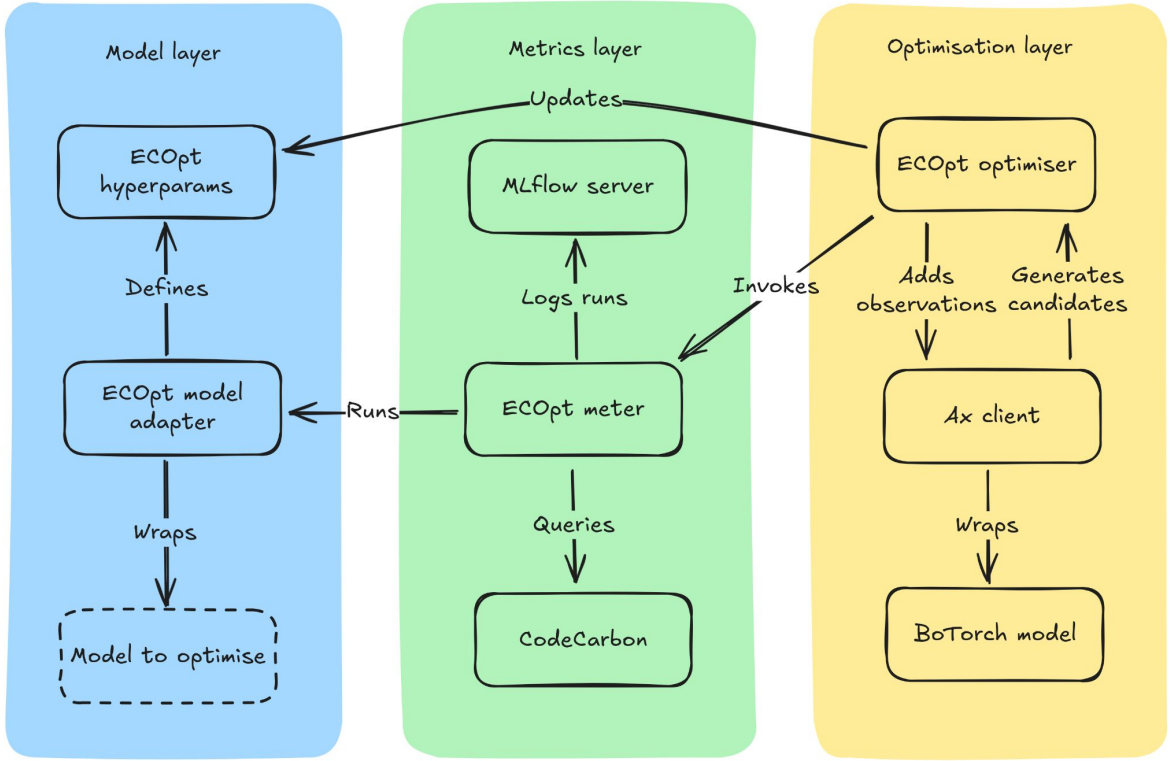
Measuring energy efficiency

Multiple hardware choices: local, HPC, virtualised

CodeCarbon as a meter

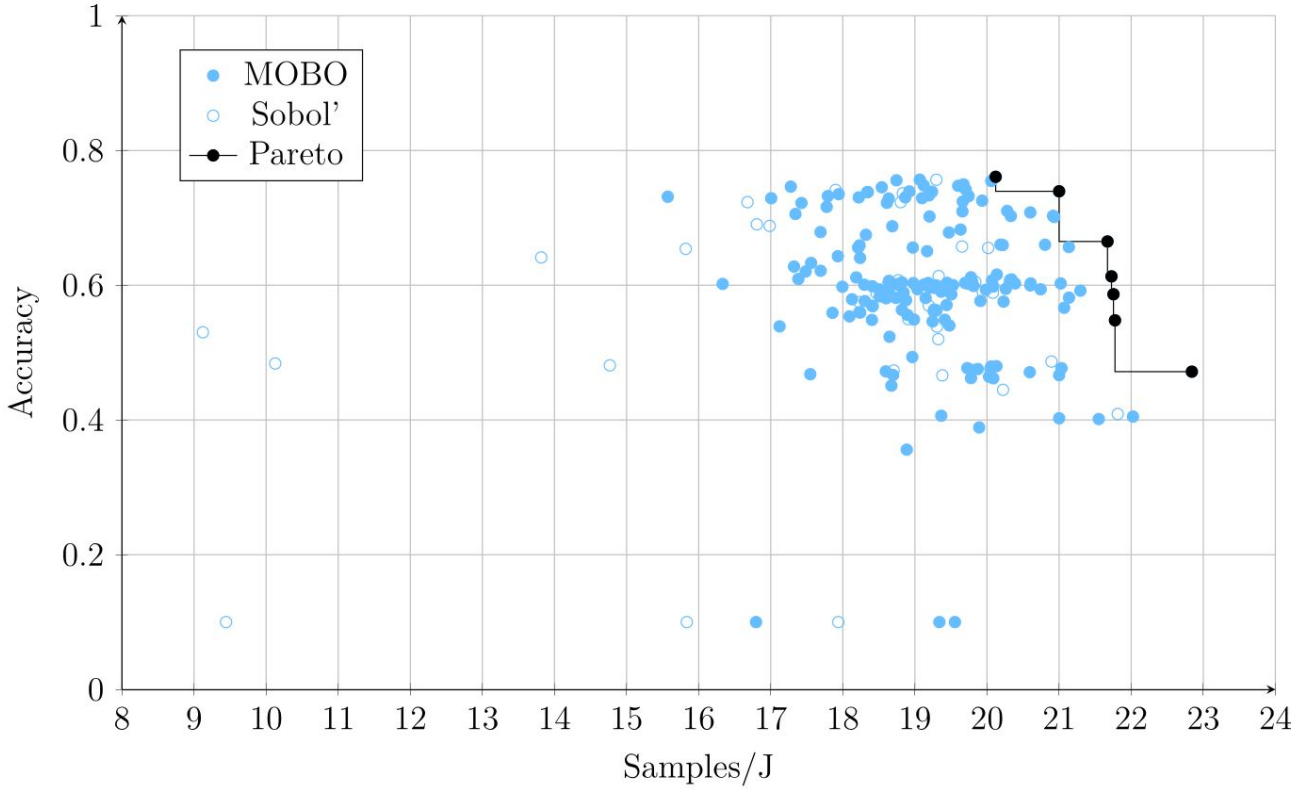
Samples per Joule as a unit

Implementation: ECOpt

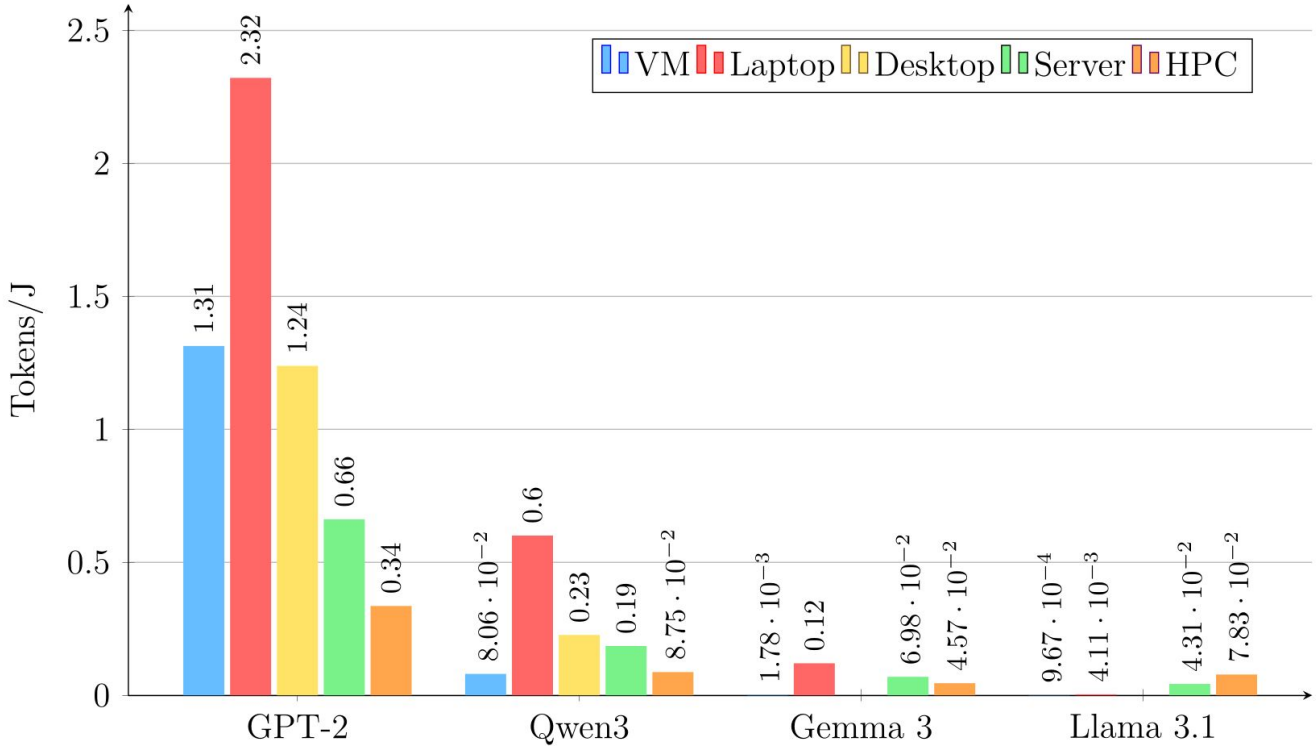


https://github.com/emileferreira/ecopt

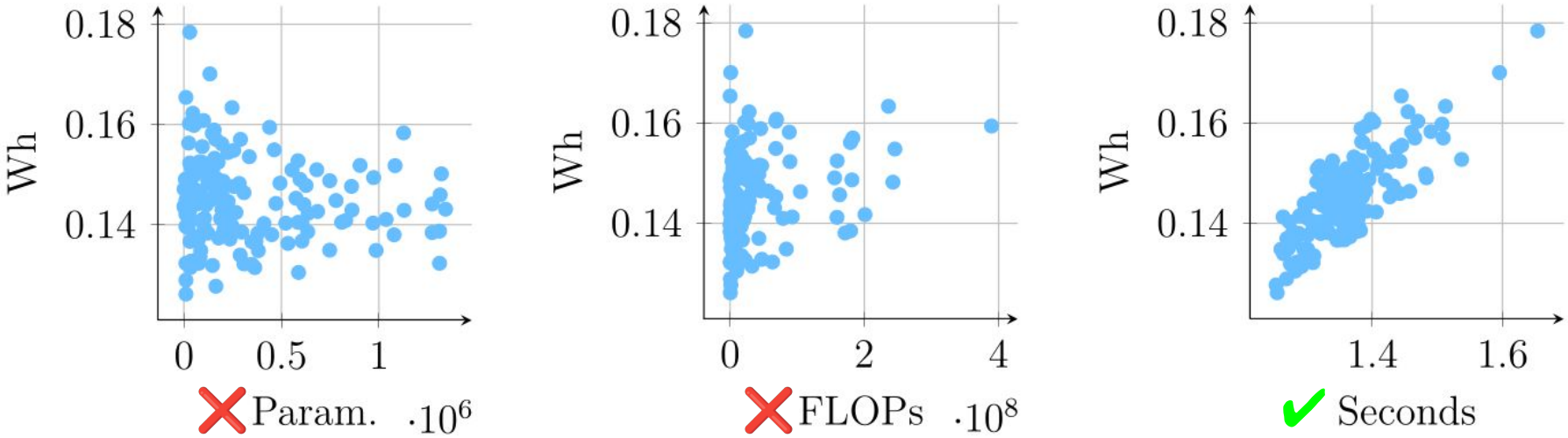
Results: discovery of empirical Pareto fronts



Results: energy metrics are consistent across hardware classes



Results: params, FLOPs are not always correlated with energy



Summary

- Performance-energy trade-off can be framed as mutli-objective optimisation
- Pareto fronts inform decisions about cost while maximising benefit
- Also help account and comply with new regulations
- We suggest and implement general procedure via ECOpt
- Parameters and FLOPs are uncorrelated with energy cons.
- Energy efficiency is relatively consistent across hardware
- **Energy metrics are worth publishing!**

Questions?

https://mlatcl.github.io/

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